

Question Paper

Exam Date & Time: 22-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

FIRST SEMESTER M.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Advanced Data Structures and Algorithms [CSS_5101]

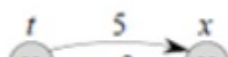
Marks: 50

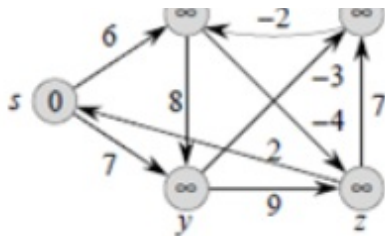
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given a real-world system where occasional costly operations occur (e.g., web browser), apply amortized reasoning to demonstrate why amortized cost gives a more realistic performance guarantee compared to worst-case bound. Consider the stack data structure implementation of the web browser operations. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Explain how a potential function based on the number of 1s in a binary counter helps derive amortized cost for INCREMENT operations. (5)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) A B-Tree of $t=3$ contains the keys: 8, 9, 10, 11, 15, 16, 17, 18, 20, 23 (already inserted) (5)
- A) Apply B-Tree deletion to remove the keys in this order: 10, 20, 8. Show the tree after each deletion and justify each case used.
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Binary heaps, binomial heaps, and Fibonacci heaps are all used to implement priority queues, but they differ significantly in structure, supported operations, and performance. Analyze the similarities and differences among these three heap types with respect to structure, operations, time complexity and real world applications. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply MAKE-SET and UNION operations to group the elements {1,2,3} using the sequence: (5)
- A) MAKE-SET(1), MAKE-SET(2), MAKE-SET(3), UNION(1,2). Show the resulting sets.
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Describe the pseudocode for the SUCCESSOR operation in a proto-vEB structure and explain how the search proceeds when the successor is inside the same cluster versus outside it. (5)
- Course Outcome: "CO3" / Cognitive Level: "Understand"
- 4) Apply the Bellman-Ford algorithm to the given weighted directed graph and determine whether a negative-weight cycle exists. Show all the steps of computation. (5)
- A)





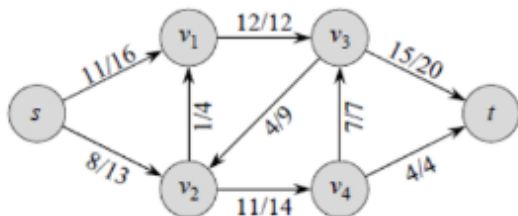
Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Apply Johnson's algorithm to the following graph and compute the shortest path matrix: $V=\{0,1,2,3\}$ and (5)
 Edges: (0,1,2), (1,2,-3), (2,3,4), (0,3,10).
 Edge details here are given in the form: (source_vertex, destination_vertex, weight).

Show all steps of computation: Bellman-Ford phase, edge reweighting, and Dijkstra phase.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 5) Apply the Ford-Fulkerson method to compute the maximum flow in the flow network shown below. (5)
 A) Show the sequence of augmenting paths, residual capacities after each iteration, and the final maximum flow value.



Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Analyze the limitations of the Ford-Fulkerson method when applied to a network with real-valued edge capacities. Explain why it may fail to terminate. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM PG-2025-2026 CSS_5101:Advanced Data Structures and Algorithms

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	10
3	3	CO 3	10
4	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM PG-2025-2026 CSS_5101:Advanced Data Structures and Algorithms

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)

3	Application	6	30(60%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)